

Multiple Choice (10 marks)

1. Which animals would most likely be helped by flood in a coastal area?
 - (a) deer
 - (b) raccoons
 - (c) alligators
 - (d) coyotes
2. How does acid rain most likely affect forests?
 - (a) Top soil loses all of its nutrients.
 - (b) Plants begin to grow deeper roots.
 - (c) Animals have more food sources.
 - (d) Trees become less healthy over time.
3. Giant redwood trees change energy from one form to another. How is energy changed by the trees?
 - (a) They change chemical energy into kinetic energy.
 - (b) They change solar energy into chemical energy.
 - (c) They change wind energy into heat energy.
 - (d) They change mechanical energy into solar energy.
4. Two layers of sedimentary rock are exposed on the side of a hill. Only one of the layers has fossils. The lack of fossils in one layer of sedimentary rock is most likely due to
 - (a) changing environments.
 - (b) volcanic activity.
 - (c) changing rates of erosion.
 - (d) evaporation of sea water.
5. Although they belong to the same family, an eagle and a pelican are different. What is one difference between them?
 - (a) their preference for eating fish
 - (b) their ability to fly
 - (c) their method of reproduction
 - (d) their method of catching food

True / False (10 marks)

Answer True or False

6. True or False: The mass of the feathers is the same as the mass of the iron nails because they both have the same volume.
7. True or False: Grass obtains its energy primarily from nutrients in the soil.
8. True or False: Moons are celestial bodies that can also be found outside the solar system.
9. True or False: A star is a natural source of light that emits energy through nuclear fusion.
10. True or False: The digestive system has layers of smooth muscle tissue that contract to move nutrients and waste through the body.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that influence the decomposition rate of various materials in nature and identify which items decompose the fastest, providing reasoning for your choices.
12. Discuss the processes involved in the transformation of sedimentary rock into metamorphic rock, and analyze the conditions necessary for this change to occur.

End of Paper